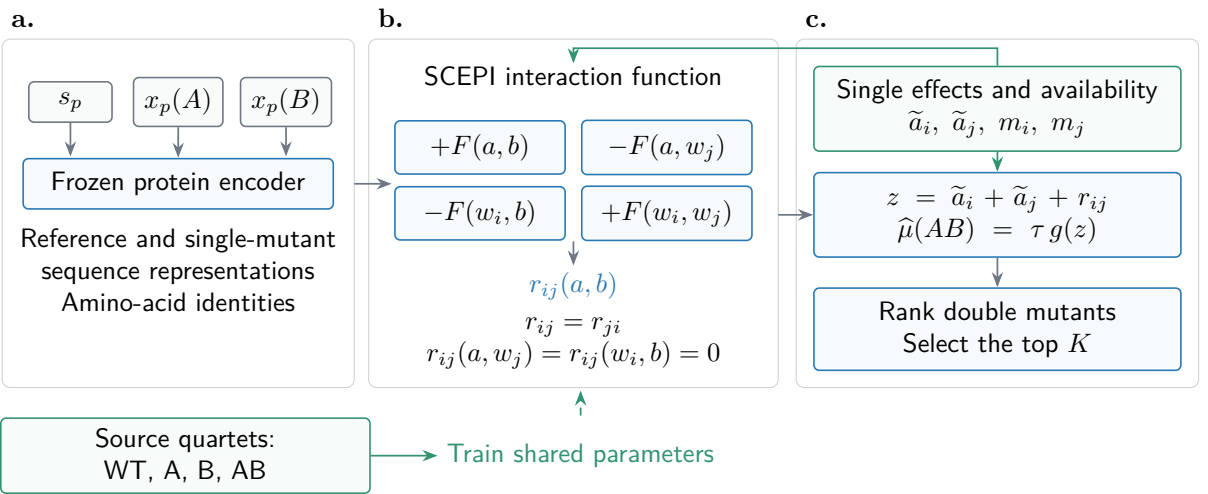
a.
b.
c.
s_p
$x_p(A)$
$x_p(B)$
Frozen protein encoder
Reference and single-mutant sequence representations
Amino-acid identities
SCEPI interaction function
$+F(a,b)$
$-F(a,w_j)$
$-F(w_i,b)$
$+F(w_i,w_j)$
$r_{ij}(a,b)$
$r_{ij} = r_{ji}$
$r_{ij}(a,w_j) = r_{ij}(w_i,b) = 0$
Single effects and availability
$\widetilde{a}_i,\ \widetilde{a}_j,\ m_i,\ m_j$
$z = \widetilde{a}_i + \widetilde{a}_j + r_{ij}$
$\widehat{\mu}(AB) = \tau\, g(z)$
Rank double mutants
Select the top K
Source quartets:
WT, A, B, AB
Train shared parameters